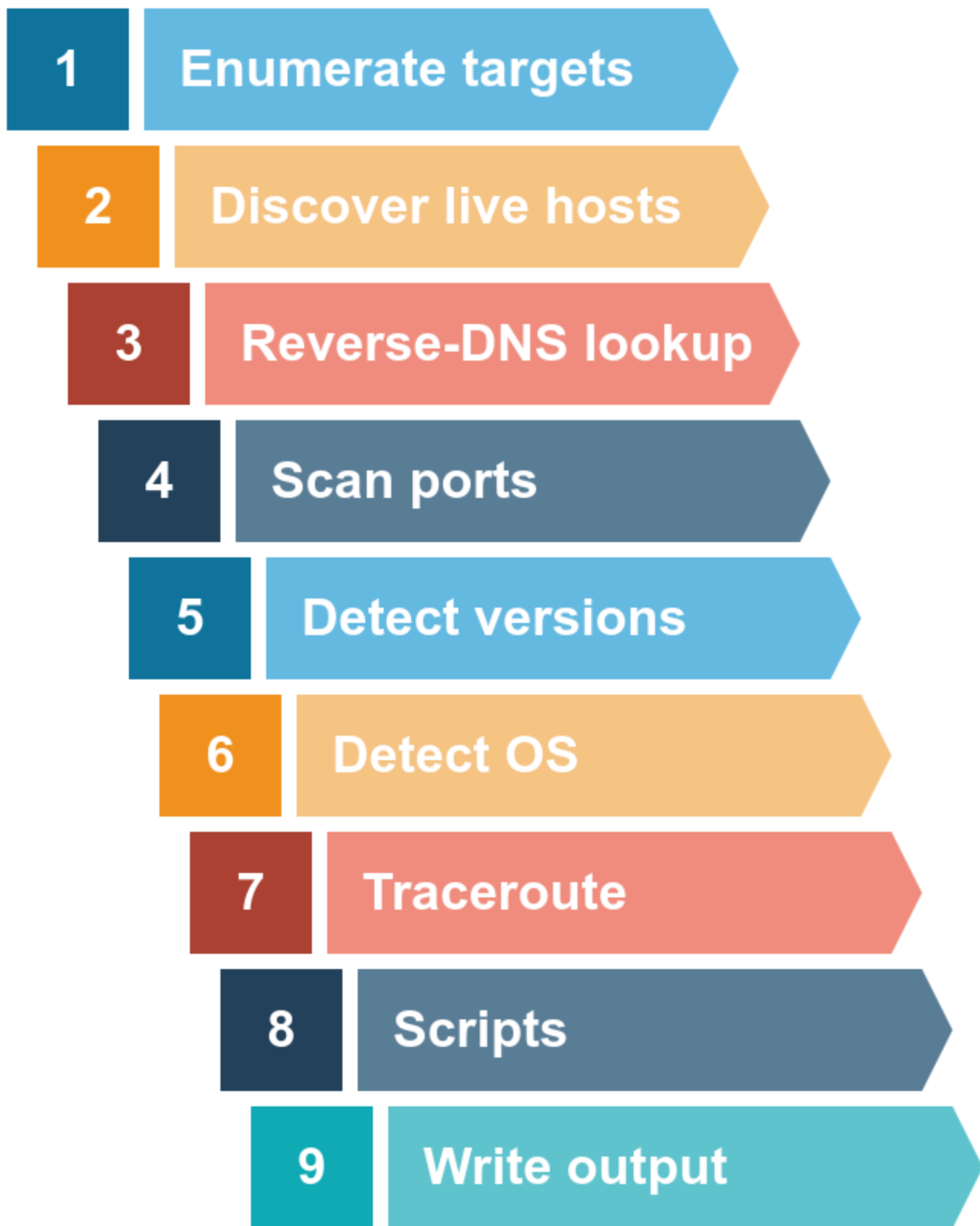


Nmap Live Host Discovery

Introduction

Nmap is an industry-standard tool for mapping networks, identifying live hosts, and discovering running services



We present the different approaches that Nmap uses to discover live hosts. In particular, we cover:

1. **ARP scan:** This scan uses ARP requests to discover live hosts
2. **ICMP scan:** This scan uses ICMP requests to identify live hosts
3. **TCP/UDP ping scan:** This scan sends packets to TCP ports and UDP ports to determine

live hosts.

Subnetworks

A *network segment* is a group of computers connected using a shared medium



a subnet, has its own IP address range and is connected to a more extensive network via a router. There might be a firewall enforcing security policies depending on each network.

- Subnets with /16 , which means that the subnet mask can be written as 255.255.0.0 . This subnet can accommodate around 65,000 hosts.
- Subnets with /24 , which indicates that the subnet mask can be expressed as 255.255.255.0 . This subnet can have around 250 hosts.

As part of active reconnaissance, we want to discover more information about a group of hosts or about a subnet. If you are on the same subnet, you would expect your scanner to use ARP Address Resolution Protocol (ARP) queries to discover live hosts

ARP is a link-layer protocol, and ARP packets are bound to their subnet

Enumerating Targets

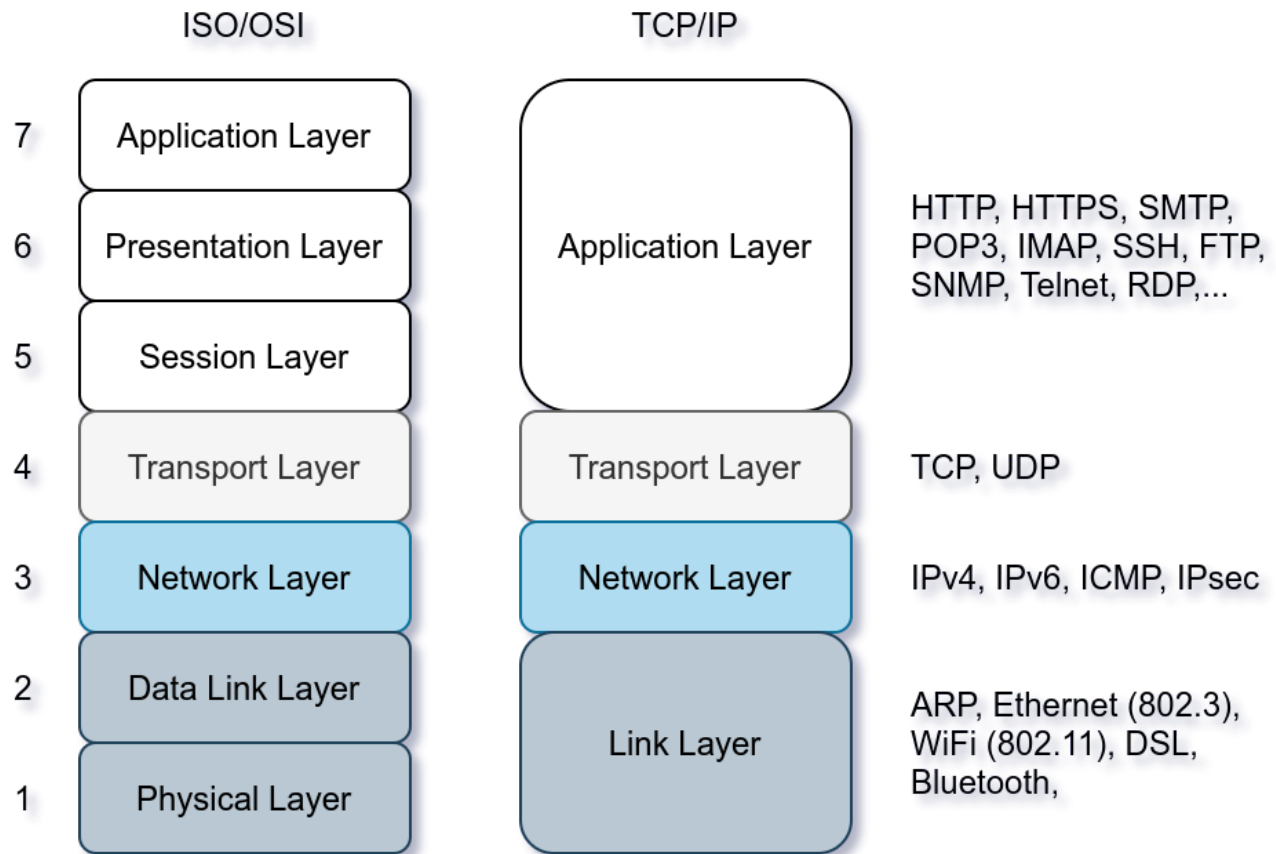
- list: `MACHINE_IP scanme.nmap.org example.com` will scan 3 IP addresses.
- range: `10.11.12.15-20` will scan 6 IP addresses: `10.11.12.15` , `10.11.12.16` ,... and `10.11.12.20` .
- subnet: `MACHINE_IP/30` will scan 4 IP addresses.

You can also provide a file as input for your list of targets, `nmap -iL list_of_hosts.txt` .

`nmap -sL TARGETS` - check the list of hosts that Nmap will scan

```
nmap scan report for 10.10.255.122
Nmap scan report for 10.10.255.123
Nmap scan report for 10.10.255.124
Nmap scan report for 10.10.255.125
Nmap done: 6400 IP addresses (0 hosts up) scanned in 0.11 seconds
root@ip-10-129-105-154:~# nmap -sL -n 10.10.12.13/29
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-16 16:54 BST
Nmap scan report for 10.10.12.8
Nmap scan report for 10.10.12.9
Nmap scan report for 10.10.12.10
Nmap scan report for 10.10.12.11
Nmap scan report for 10.10.12.12
Nmap scan report for 10.10.12.13
Nmap scan report for 10.10.12.14
Nmap scan report for 10.10.12.15
Nmap done: 8 IP addresses (0 hosts up) scanned in 0.00 seconds
```

Discovering Live Hosts



- ARP from Link Layer
- ICMP from the Network Layer
- TCP from the Transport Layer
- UDP from the Transport Layer

ping a system on the same subnet, an ARP query should precede the ICMP Echo.

Nmap Host Discovery Using ARP

1. When a *privileged* user tries to scan targets on a local network (Ethernet), Nmap uses *ARP requests*. A privileged user is `root` or a user who belongs to `sudoers` and can run `sudo`.
2. When a *privileged* user tries to scan targets outside the local network, Nmap uses ICMP echo requests, TCP ACK (Acknowledge) to port 80, TCP SYN (Synchronize) to port 443, and ICMP timestamp request.
3. When an *unprivileged* user tries to scan targets outside the local network, Nmap resorts to a TCP 3-way handshake by sending SYN packets to ports 80 and 443.

Nmap to discover online hosts without port-scanning the live systems:

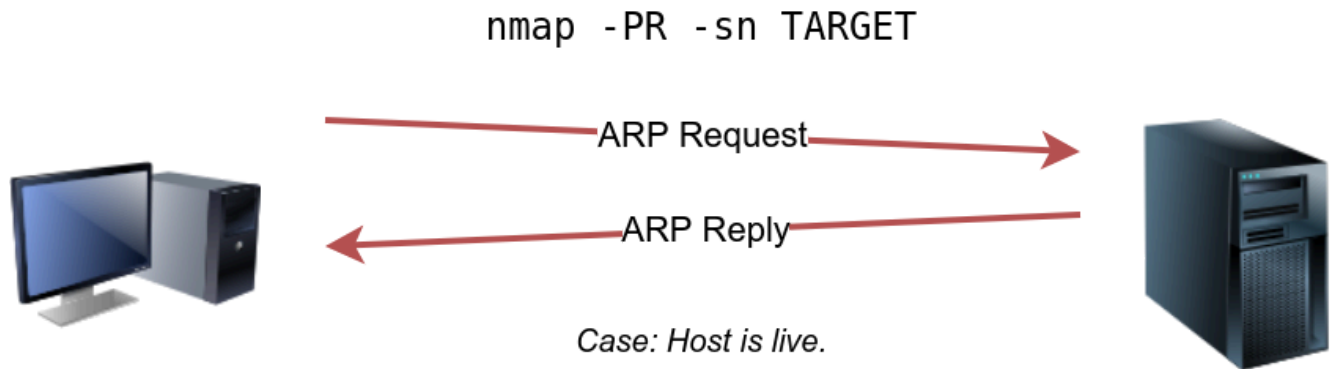
```
nmap -sn TARGETS
```

Nmap only to perform an ARP scan without port-scanning:

```
nmap -PR -sn TARGETS
```

to discover all the live systems on the same subnet as our target machine:

```
nmap -PR -sn MACHINE_IP/24
```



```
arp-scan
```

popular choice is `arp-scan --localnet` or simply `arp-scan -l`. This command will send ARP queries to all valid IP addresses on your local networks

`sudo arp-scan -I eth0 -l` will send ARP queries for all valid IP addresses on the `eth0` interface

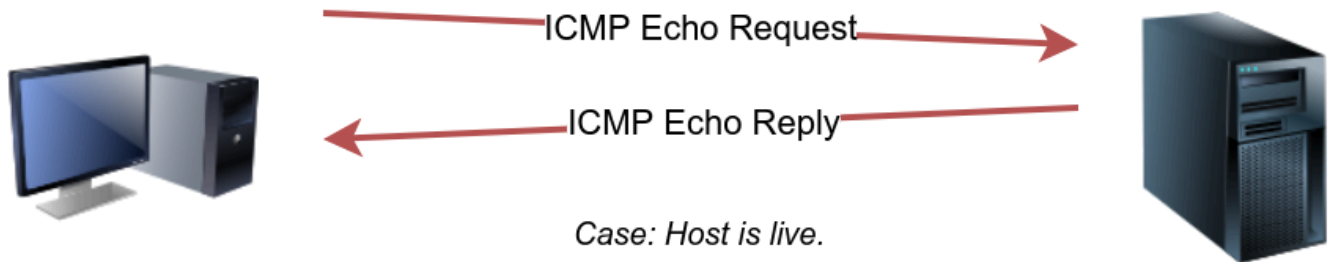
```
pentester@TryHackMe$ sudo arp-scan 10.10.210.6/24
Interface: eth0, datalink type: EN10MB (Ethernet)
WARNING: host part of 10.10.210.6/24 is non-zero
Starting arp-scan 1.9 with 256 hosts (http://www.nta-monitor.com/tools/arp-scan/)
10.10.210.75    02:83:75:3a:f2:89    (Unknown)
10.10.210.100  02:63:d0:1b:2d:cd    (Unknown)
10.10.210.165  02:59:79:4f:17:b7    (Unknown)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9: 256 hosts scanned in 2.726 seconds (93.91 hosts/sec). 3 respon
```

Nmap Host Discovery Using ICMP

To use ICMP echo request to discover live hosts, add the option `-PE`. (Remember to add `-sn` if you don't want to follow that with a port scan.)

nmap -PE -sn TARGET



nmap -PE -sn MACHINE_IP/24

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 10:16 BST
Nmap scan report for ip-10-10-68-50.eu-west-1.compute.internal (10.10.68.50)
Host is up (0.00017s latency).
MAC Address: 02:95:36:71:5B:87 (Unknown)
Nmap scan report for ip-10-10-68-52.eu-west-1.compute.internal (10.10.68.52)
Host is up (0.00017s latency).
MAC Address: 02:48:E8:BF:78:E7 (Unknown)
Nmap scan report for ip-10-10-68-77.eu-west-1.compute.internal (10.10.68.77)
Host is up (-0.100s latency).
MAC Address: 02:0F:0A:1D:76:35 (Unknown)
Nmap scan report for ip-10-10-68-110.eu-west-1.compute.internal (10.10.68.110)
Host is up (-0.10s latency).
MAC Address: 02:6B:50:E9:C2:91 (Unknown)
Nmap scan report for ip-10-10-68-140.eu-west-1.compute.internal (10.10.68.140)
Host is up (0.00021s latency).
MAC Address: 02:58:59:63:0B:6B (Unknown)
Nmap scan report for ip-10-10-68-142.eu-west-1.compute.internal (10.10.68.142)
Host is up (0.00016s latency).
MAC Address: 02:C6:41:51:0A:0F (Unknown)
Nmap scan report for ip-10-10-68-220.eu-west-1.compute.internal (10.10.68.220)
Host is up (0.00026s latency).
MAC Address: 02:25:3F:DB:EE:0B (Unknown)
Nmap scan report for ip-10-10-68-222.eu-west-1.compute.internal (10.10.68.222)
Host is up (0.00025s latency).
MAC Address: 02:28:B1:2E:B0:1B (Unknown)
Nmap done: 256 IP addresses (8 hosts up) scanned in 2.11 seconds
```

We will repeat the scan above; however, this time we will scan from a system on a different subnet. The results are similar but without the MAC addresses.

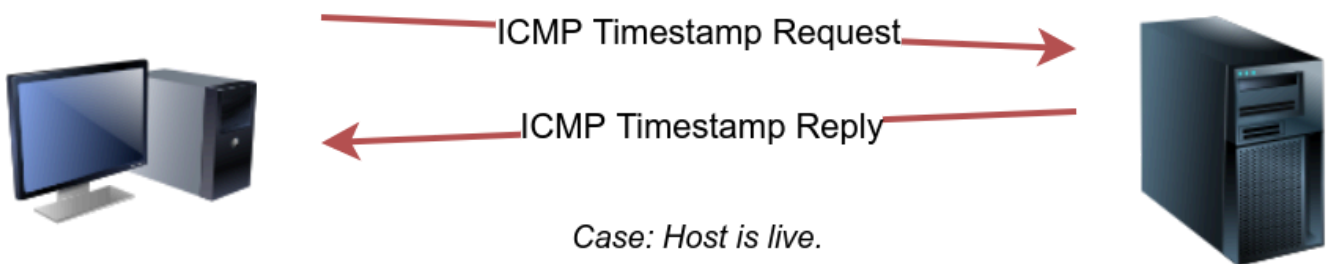
```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:16 EEST
Nmap scan report for 10.10.68.50
Host is up (0.12s latency).
Nmap scan report for 10.10.68.52
Host is up (0.12s latency).
Nmap scan report for 10.10.68.77
Host is up (0.11s latency).
Nmap scan report for 10.10.68.110
Host is up (0.11s latency).
Nmap scan report for 10.10.68.140
Host is up (0.11s latency).
Nmap scan report for 10.10.68.142
Host is up (0.11s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap scan report for 10.10.68.222
Host is up (0.11s latency).
Nmap done: 256 IP addresses (8 hosts up) scanned in 8.26 seconds
```

Adding the `-PP` option tells Nmap to use ICMP timestamp requests.

`nmap -PP -sn TARGET`



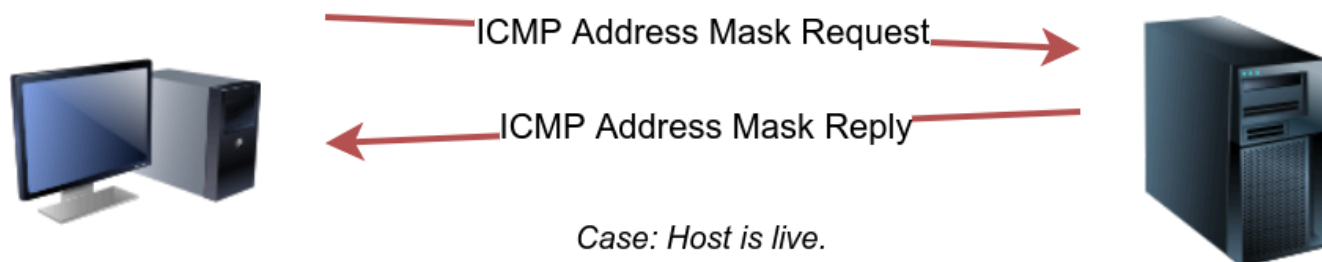

```
nmap -PP -sn MACHINE_IP/24
```

```
pentester@TryHackMe$ sudo nmap -PP -sn 10.10.68.220/24
```

```
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:06 EEST
Nmap scan report for 10.10.68.50
Host is up (0.13s latency).
Nmap scan report for 10.10.68.52
Host is up (0.25s latency).
Nmap scan report for 10.10.68.77
Host is up (0.14s latency).
Nmap scan report for 10.10.68.110
Host is up (0.14s latency).
Nmap scan report for 10.10.68.140
Host is up (0.15s latency).
Nmap scan report for 10.10.68.209
Host is up (0.14s latency).
Nmap scan report for 10.10.68.220
Host is up (0.14s latency).
Nmap scan report for 10.10.68.222
Host is up (0.14s latency).
Nmap done: 256 IP addresses (8 hosts up) scanned in 10.93 seconds
```

Nmap uses address mask queries (ICMP Type 17) and checks whether it gets an address mask reply (ICMP Type 18). This scan can be enabled with the option `-PM`

```
nmap -PM -sn TARGET
```



```
nmap -PM -sn MACHINE_IP/24
```

```
Pentester Terminal

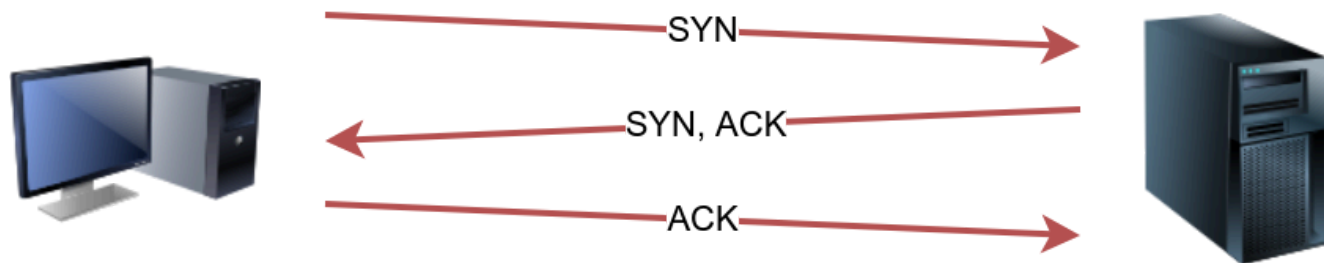
pentester@TryHackMe$ sudo nmap -PM -sn 10.10.68.220/24

Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:13 EEST
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.17 seconds
```

Nmap Host Discovery Using TCP and UDP

TCP SYN Ping

TCP 3-Way Handshake

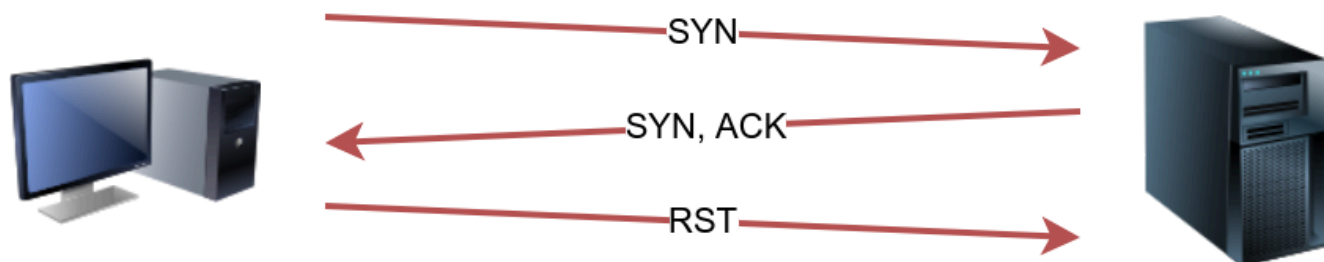


Case: TCP port is open.

Nmap to use TCP SYN ping, you can do so via the option `-PS` followed by the port number, range, list, or a combination of them

`-PS21` will target port 21, while `-PS21-25` will target ports 21, 22, 23, 24, and 25. Finally `-PS80,443,8080` will target the three ports 80, 443, and 8080.

nmap -PS -sn TARGET



Case: TCP port is open.

```
nmap -PS -sn MACHINE_IP/24
```

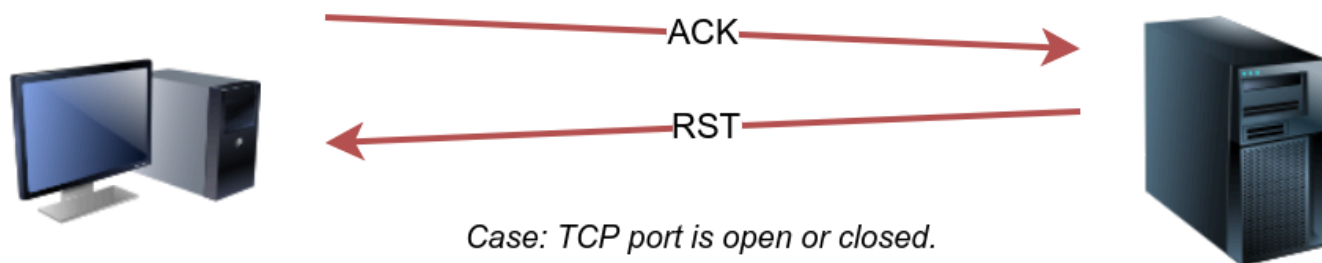
```
pentester@TryHackMe$ sudo nmap -PS -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.16s latency).
Nmap scan report for 10.10.68.125
Host is up (0.089s latency).
Nmap scan report for 10.10.68.134
Host is up (0.13s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 17.38 seconds
```

TCP ACK Ping

run Nmap as a privileged user to accomplish this. If you try it as an unprivileged user, Nmap will attempt a 3-way handshake.

The syntax is similar to TCP SYN ping. `-PA` should be followed by a port number, a port number range, a list, or a combination of them. For example, consider `-PA21` , `-PA21-25` and `-PA80,443,8080` . If no port is specified, port 80 will be used.

`nmap -PA -sn TARGET`



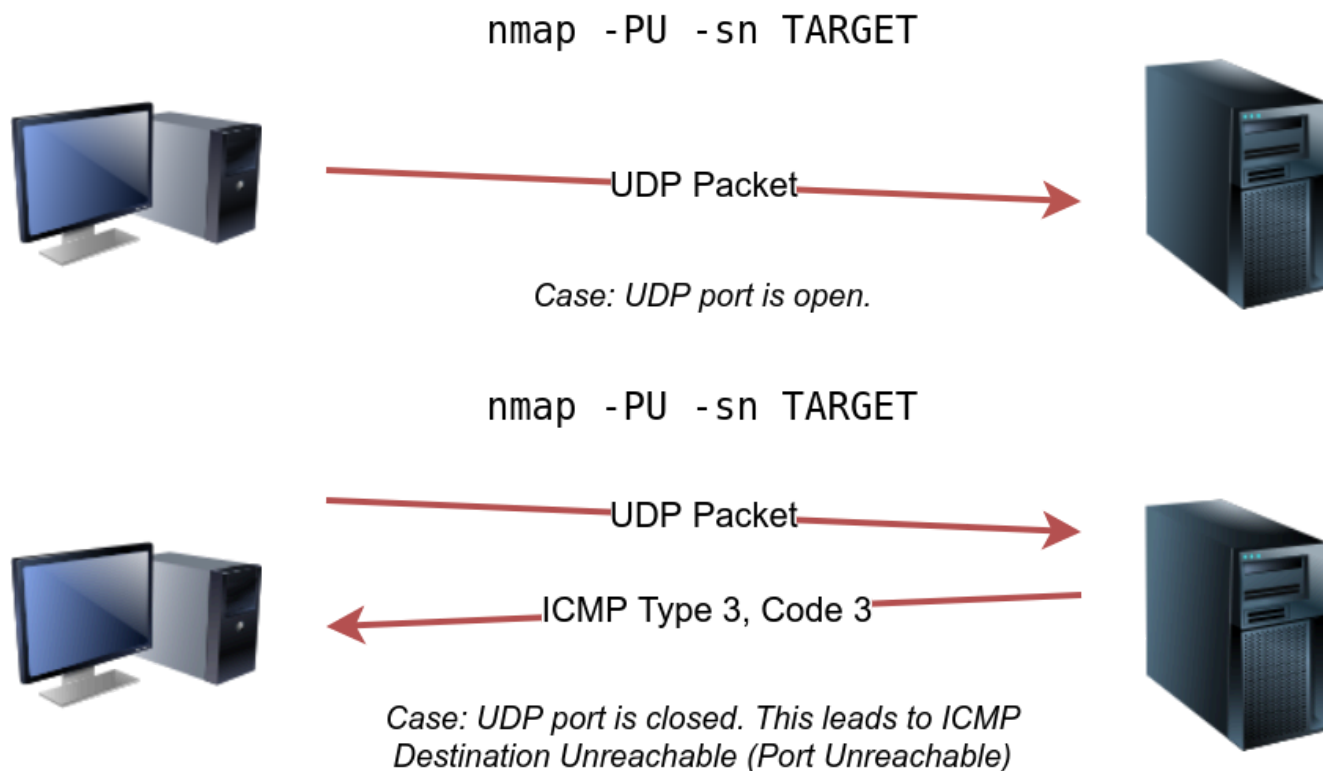
```
sudo nmap -PA -sn MACHINE_IP/24
```

```
pentester@TryHackMe$ sudo nmap -PA -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:46 EEST
Nmap scan report for 10.10.68.52
Host is up (0.11s latency).
Nmap scan report for 10.10.68.121
Host is up (0.12s latency).
Nmap scan report for 10.10.68.125
Host is up (0.10s latency).
Nmap scan report for 10.10.68.134
Host is up (0.10s latency).
Nmap scan report for 10.10.68.220
Host is up (0.10s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 29.89 seconds
```

UDP Ping

UDP packet to an open port is not expected to lead to any reply

if we send a UDP packet to a closed UDP port, we expect to get an ICMP port unreachable packet; this indicates that the target system is up and available.



```
sudo nmap -PU -sn 10.10.68.220/24
```

```
Pentester Terminal

pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```

Masscan

- `masscan MACHINE_IP/24 -p443`
- `masscan MACHINE_IP/24 -p80,443`
- `masscan MACHINE_IP/24 -p22-25`
- `masscan MACHINE_IP/24 --top-ports 100`

Using Reverse-DNS Lookup

Nmap's default behaviour is to use reverse-DNS online hosts. Because the hostnames can reveal a lot, this can be a helpful step. However, if you don't want to send such DNS queries, you use `-n` to skip this step.

By default, Nmap will look up online hosts; however, you can use the option `-R` to query the DNS server even for offline hosts. If you want to use a specific DNS server, you can add the `--dns-servers DNS_SERVER` option

Summary

Scan Type	Example Command
ARP Scan	<code>sudo nmap -PR -sn MACHINE_IP/24</code>
ICMP Echo Scan	<code>sudo nmap -PE -sn MACHINE_IP/24</code>
ICMP Timestamp Scan	<code>sudo nmap -PP -sn MACHINE_IP/24</code>
ICMP Address Mask Scan	<code>sudo nmap -PM -sn MACHINE_IP/24</code>
TCP SYN Ping Scan	<code>sudo nmap -PS22,80,443 -sn MACHINE_IP/30</code>
TCP ACK Ping Scan	<code>sudo nmap -PA22,80,443 -sn MACHINE_IP/30</code>
UDP Ping Scan	<code>sudo nmap -PU53,161,162 -sn MACHINE_IP/30</code>

Remember to add `-sn` if you are only interested in host discovery without port-scanning. Omitting `-sn` will let Nmap default to scanning live hosts for ports.

Option	Purpose
<code>-n</code>	no DNS lookup
<code>-R</code>	reverse-DNS lookup for all hosts
<code>-sn</code>	host discovery only